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PROJECTS

Redis-Backed Message Queue System — Node.js, BullMQ, Redis

Mar 2026

- **Abstract:** Built a message queue system using Node.js with BullMQ and Redis, implementing a producer-consumer architecture for asynchronous task processing. The system manages background jobs through a Redis-backed queue with dedicated worker processes, enabling scalable task execution outside the request-response cycle.
- Implemented producer-consumer architecture pushing background jobs to Redis and consuming them via dedicated worker processes, decoupling task execution from API calls.
- Gained deep understanding of job state management, queue resilience, and handling long-running tasks asynchronously to maintain API responsiveness.
- Implemented advanced queue features including job retries, priority management, task delays, and comprehensive state tracking (Waiting, Active, Completed).
- Applied Redis as a robust state management layer, leveraging its performance and persistence for reliable queue operations at scale.
- **Tech Stack:** Node.js, BullMQ, Redis, ioredis, JavaScript

ThermoSafe — React, Firebase, ESP32, IoT

Jan 2026

- Built an end-to-end IoT solution to monitor and protect temperature-sensitive goods during storage and transportation.
- Implemented real-time temperature tracking and instant alerts to reduce losses in cold-chain logistics.
- Designed for scalability and practical deployment, supporting applications like vaccine transportation, pharmaceutical storage.
- Integrated ESP32 microcontroller with DS18B20 sensors to reliably collect temperature data and sync with Firebase for cloud-based monitoring.
- Deployment Link: <https://thermosafe.up.railway.app>

Travel Tinder — React.js, Node.js, Express.js, Firebase

Apr 2025

- Developed a full-stack web application to connect solo travelers using a swipe-based travel companion matching system.
- Implemented Firebase-based authentication including OTP login and secure profile handling with SSO.
- Built RESTful APIs following MVC architecture to handle matching logic, user data, and real-time interactions.
- Designed trip planning and in-app chat features to enable collaboration between matched users.
- Integrated a subscription-based model (Basic, Pro, Premium) to unlock premium features and support monetization.

Crop Disease Detection and Prediction (CropVit) — Vision Transformer, PyTorch, Flask, MongoDB Dec 2024

- Developed a crop disease detection system using Vision Transformer models for major crops including Orange, Cotton, Soybean, and Potato.
- Achieved 98.80% classification accuracy with minimal loss through optimized training and preprocessing.
- Created a Flask-based web application providing real-time disease detection and solution recommendations to support data-driven decisions for farmers.

Efficient Multi-Node Cluster Deployment and Spark Job Execution

Nov 2024

- **Abstract:** Developed a robust framework for deploying and managing multi-node Spark clusters, providing a scalable foundation for distributed big data analytics, stream processing, and real-time decision-making systems.
- Implemented distributed computing infrastructure supporting parallel job execution across multiple nodes with fault tolerance mechanisms.
- Optimized cluster configuration and job scheduling to achieve efficient resource utilization and improved data processing throughput.
- Designed for horizontal scalability, enabling seamless addition of compute nodes to handle growing data volumes and workload demands.
- **Tech Stack:** Apache Spark, Hadoop, Python, YARN, SSH

DigiAttendance — Python, Tkinter, MySQL, Git-GitHub, Excel Integration

Nov 2023

- Developed a secure attendance management system for educational institutions using QR code scanning and location verification.
- Implemented authentication mechanisms to prevent proxy attendance and ensure data integrity.
- Enabled real-time attendance tracking and reporting with MySQL database integration and Excel exports.

EXPERIENCE

Open Source Maintainer — The RoboTech Forum

Feb 2026 - Present

- Serve as a core maintainer for the official website repository, ensuring code stability and optimal performance.
- Review, test, and merge community pull requests while managing issue trackers and collaborating with open-source contributors.
- Develop and optimize front-end features and user interfaces utilizing **JavaScript, HTML, and CSS**.

Technical Intern — ITRNITY INFOTECH

Jan 2025 – Apr 2025

- Worked on the design and development of **Travel Tinder**, a full-stack web application to connect solo travelers using a swipe-based matching system.
- Developed RESTful APIs using Node.js and Express.js following MVC architecture for user management, matching logic, and subscriptions.
- Gained hands-on experience with cloud computing fundamentals including deployment concepts, server management, and scalability considerations.
- Learned CI/CD, containerization basics, and version control workflows using Git and GitHub.

EDUCATION

B.Tech. in Information Technology

Government College of Engineering, Amravati

Pursuing: 2028

CGPA: 8.0 / 10

Diploma in Computer Engineering

Government Polytechnic Nagpur

Completed

Percentage: 92.50

SKILLS SUMMARY

- **Languages:** Python, SQL, Java,C,JavaScript,Shell
- **Frameworks:** React, Node, Tailwind,
- **Developer Tools:** VS Code, Android Studio, GitHub, Firebase, Google Colab, Kiro, Antigravity
- **Cloud Platforms:** Google Cloud

CERTIFICATIONS & ACHIEVEMENTS

- **Google Cloud Arcade Program:** Completed GCP labs and gain the Cloud knowledge .
- **Fundamentals of Accelerated Computing with CUDA C/C++** – NVIDIA
- **Postman API Fundamentals Student Expert Badge** – Postman
- **Fundamentals of Cloud Computing** – Itrnity Infotech
- **SIH Mentor (2025):** Collaborated with a team to build a cloud-based practice management and nutrient analysis system for Ayurvedic dietitians.
- **Published Research Paper:** State-level paper presentation competition, Government Polytechnic Yavatmal.